EmberIQ

A Project - IV

Submitted in partial fulfillment of the requirement for the award of Degree of Bachelor of Technology in Computer Science & Engineering-AIML

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(A UGC Autonomous Institute, Affiliated to RGPV)

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Project-IV entitled

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For the partial fulfillment for the award of the Bachelor of Technology (Computer Science & Engineering-AIML) Degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal.

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CERTIFICATE

This is to certify that Project -IV entitled

"EmberIQ"

has been successfully completed by the following students

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in partial fulfillment for the award of the Bachelor of Technology (Computer Science & Engineerin-AIML) Degree by Rajiv Gandhi Proudyogiki Vishwavidhyalaya, Bhopal during the academic year 2025-26 under our guidance.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
JWT	JSON Web Token
UI	User Interface
UX	User Experience
CSS	Cascading Style Sheets
HTML	Hyper Text Markup Language
JSON	JavaScript Object Notation
CRUD	Create, Read, Update, Delete
CI/CD	Continuous Integration / Continuous Deployment
WebRTC	Web Real-Time Communication
DB	Database
NoSQL	Non-relational Structured Query Language
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
IDE	Integrated Development Environment
DOM	Document Object Model
REST	Representational State Transfer
API Gateway	Interface for managing APIs
MVC	Model View Controller

ABSTRACT

EmberIQ is a comprehensive, full-stack remote interview platform designed to streamline and modernize the technical hiring process in a remote-first environment. With the increasing reliance on online interviews, existing solutions often suffer from fragmented workflows, lack of real-time collaboration, inefficient evaluation methods, and security vulnerabilities. These limitations create a suboptimal experience for both interviewers and candidates.

The proposed system addresses these challenges by integrating video conferencing, real-time collaborative coding, structured interview management, and AI-assisted evaluation into a single unified platform. EmberIQ enables seamless interaction between interviewers and candidates through secure video and audio communication, live code editing with multi-language support, and built-in chat and screen sharing features.

The platform incorporates a real-time coding environment powered by Socket.IO, allowing interviewers to observe and interact with candidates' code as it is written. Additionally, AI-assisted features provide coding suggestions, performance insights, and help reduce bias in evaluation. A centralized dashboard facilitates interview scheduling, candidate tracking, and structured feedback collection through scorecards and notes.

Security and privacy are ensured through the implementation of JWT-based authentication, encrypted data handling, and role-based access control, protecting sensitive interview data from unauthorized access. The system is built using modern technologies including React.js, Node.js, Express.js, MongoDB, and WebRTC, ensuring scalability, performance, and a responsive user experience.

Overall, EmberIQ provides a robust, efficient, and secure solution for conducting technical interviews, improving the hiring workflow, enhancing collaboration, and delivering a consistent and professional interview experience.

Keywords:

Machine Learning, Student Performance, Real-time Collaboration, Audio communication, Live Code Editing, Educational Data Mining.

CHAPTER 1

INTRODUCTION

1.1 Background and Context

In recent years, the global shift toward digital transformation has significantly changed the way organizations conduct recruitment processes. With the rise of remote work culture and geographically distributed teams, companies increasingly rely on online platforms to conduct technical interviews. This transition has accelerated especially after global events such as the COVID-19 pandemic, which made remote hiring not just an option but a necessity.

However, despite the availability of tools like video conferencing platforms and online coding websites, the current interview ecosystem remains highly fragmented. Interviewers are often required to switch between multiple applications such as video calling tools, coding platforms, and evaluation sheets. This fragmented approach not only reduces efficiency but also creates confusion and inconsistency in the interview process.

Another major challenge in existing systems is the lack of real-time collaboration. Many platforms do not provide a seamless environment where candidates can write, execute, and explain their code while interviewers observe and interact simultaneously. This limitation affects the quality of technical assessment and reduces the effectiveness of the interview.

Furthermore, the evaluation process is often manual and unstructured. Interviewers typically rely on separate documents or memory to assess candidates, which may lead to biased or inconsistent decisions. There is also a growing concern regarding data security and privacy, as sensitive candidate information and interview data are not always handled securely across different platforms.

In this context, there is a strong need for a unified, secure, and efficient system that can handle all aspects of technical interviews within a single platform. EmberIQ is developed to address these challenges by providing an integrated solution that combines video communication, real-time coding, interview management, and AI-assisted evaluation in one cohesive system.

By leveraging modern web technologies and real-time communication frameworks, EmberIQ aims to improve the overall interview experience for both interviewers and candidates while ensuring scalability, security, and ease of use.

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1.2 Importance and Relevance

- EmberIQ is highly important in today's digital hiring environment where companies are increasingly shifting towards remote interviews. Traditional online interview processes rely on multiple tools such as video platforms, coding environments, and separate evaluation systems, which creates inefficiency and confusion.
- The platform provides a unified solution by integrating video communication, real-time coding, and candidate evaluation into a single system. This eliminates the need to switch between different tools and makes the interview process more organized and efficient.
- EmberIQ significantly improves interview efficiency by allowing interviewers to conduct interviews, monitor coding activities, and provide feedback within one interface. This reduces time consumption and enhances productivity.
- It also enhances the candidate experience by offering a smooth, clear, and structured interview workflow. Candidates can interact, code, and communicate in real time without facing technical difficulties or platform-related issues.
- The project introduces a structured evaluation system that includes scorecards, real-time notes, and centralized feedback storage. This helps interviewers make better and more accurate hiring decisions based on proper data.
- Security and privacy are important aspects of the platform. EmberIQ ensures secure authentication, encrypted communication, and role-based access control, which protects sensitive candidate information and maintains system integrity.
- The project is highly relevant from a technological perspective as it uses modern tools like React.js, Node.js, WebRTC, and Socket.IO. These technologies make the platform scalable, responsive, and suitable for real-world applications.
- EmberIQ also supports remote and flexible hiring, allowing companies to conduct interviews from anywhere without geographical limitations. This makes the system useful for startups, companies, and educational institutions.
- Overall, EmberIQ plays a significant role in improving the efficiency, security, and experience of remote technical interviews, making it a valuable solution in the modern hiring ecosystem.

1.2.1 Benefits for Individual Users

- EmberIQ provides a simple and unified platform where users can attend interviews, write code, and communicate without switching between multiple tools. This makes the overall process smooth and reduces confusion during interviews.
- It improves the user experience by offering real-time interaction, clear workflow, and an easy-to-use interface. Candidates can focus better on their performance instead of dealing with technical issues or complex platforms.
- The platform ensures data security and privacy through secure login, encrypted sessions, and controlled access. This gives users confidence that their personal information and interview data are safe.

1.2.2 Benefits for Developers

- EmberIQ provides a well-structured architecture using modern technologies like React, Node.js, and MongoDB. This makes it easier for developers to build, integrate, and maintain different modules efficiently
- The use of WebRTC and Socket.IO enables developers to implement real-time communication and collaboration features. The system is scalable, allowing developers to extend functionalities and handle more users easily.

1.2.3 Real-World Context

EmberIQ is designed based on real-world hiring challenges faced by companies during remote interviews. Many organizations use separate tools for video calls, coding tests, and evaluation, which creates inefficiency. This platform solves that problem by providing all features in one system. It helps companies conduct interviews more smoothly and professionally without technical issues.

Therefore, EmberIQ is highly relevant in today's remote hiring and digital recruitment environment.

1.3 Scope and Limitations

1.3.1 Scope

EmberIQ is designed as a full-stack web application targeting companies, recruiters, students, and developers who require an efficient, integrated, and user-friendly platform for conducting remote technical interviews without relying on multiple disconnected tools or complex systems.

Core Modules

- **Live Interview Dashboard:** Real-time video and audio interview system using WebRTC with secure communication, chat support, screen sharing, and smooth interaction between interviewer and candidate.
- **Real-Time Code Editor:** Collaborative coding environment with support for multiple programming languages, syntax highlighting, optional code execution, and live monitoring by interviewers.
- **Interview Management System:** Dashboard for scheduling interviews, assigning interviewers, managing interview rounds, and maintaining a structured workflow for the hiring process.
- **Candidate Evaluation Module:** Scorecard-based evaluation system with real-time interviewer notes, performance tracking, and centralized feedback storage for better decision-making.
- **AI Interview Assistant:** AI-powered support system providing coding hints, explanations, automatic code analysis, and intelligent suggestions to assist interviewers during interviews.
- **Communication and Chat Module:** Built-in messaging system for real-time communication between interviewer and candidate during the interview session.
- **Security and Authentication Module:** Secure login system using JWT, encrypted sessions, and role-based access control for interviewers, candidates, and administrators.
- **Data Storage and Management:** MongoDB-based storage system for managing user data, interview records, coding responses, and feedback securely and efficiently.

1.3.2 Limitations

- **Web-Only Platform:** EmberIQ is currently available only as a web application and does not provide native Android or iOS mobile applications.
- **Dependency on External Services**: Core features rely on technologies like WebRTC and Socket.IO. Any service disruption or network issue can affect real-time communication and interview functionality.
- **No Offline Mode:** The platform requires an active internet connection for video calling.

1.4 Future Enhancements

- **Browser Extension:** Development of a browser extension to enable real-time interview assistance, quick access to coding tools, and improved user interaction during live sessions.
- **Mobile Application:** Creation of a native mobile application using React Native to allow users to attend interviews, receive notifications, and manage schedules on mobile devices.
- **Advanced AI Integration:** Implementation of more advanced AI features such as automated candidate evaluation, performance analysis, and intelligent interview recommendations.
- **Password less Authentication:** Integration of modern authentication methods like WebAuth or Passkeys to enhance security and provide a seamless login experience.
- **Team Collaboration Features:** Introduction of team-based workspaces with role-based access control, enabling multiple interviewers and admins to collaborate efficiently.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Review

Remote interview platforms exist at the intersection of real-time communication, collaborative coding environments, and web-based system design. A significant amount of research and development has focused on individual components such as video conferencing systems, online coding platforms, and digital evaluation tools. However, most existing solutions address these components separately rather than as a unified system. Various platforms like Zoom and Google Meet provide reliable video communication but lack integrated coding environments and structured interview workflows. Similarly, coding platforms such as HackerRank and LeetCode support programming assessments but do not offer real-time interviewer–candidate interaction through video and collaboration tools.

2.1.1 Evolution of Remote Interview System

Early approaches to technical interviews were conducted offline or through basic video communication tools, where interviewers relied on face-to-face interaction or simple online meetings. These methods lacked proper structure, real-time coding capabilities, and integrated evaluation systems, making the process less efficient and harder to manage.

With the advancement of online platforms, tools like Zoom and Google Meet introduced remote communication, allowing interviews to be conducted virtually. However, these platforms were not specifically designed for technical interviews and required additional tools for coding and evaluation. Later, coding platforms such as HackerRank and LeetCode provided environments for programming assessments, but they operated separately from communication tools. This created

2.1.2 Research and Case Studies

It includes key studies showing that technologies like WebRTC and WebSockets enable real-time communication and collaboration, while integrated coding platforms improve candidate assessment. It also highlights that AI-assisted systems enhance evaluation accuracy, and modern remote hiring platforms provide greater efficiency and a better user experience.

Study / Project	Authors	Key Findings
Real-Time Communication Systems	Singh et al. (2019)	Demonstrated that WebRTC enables low-latency video for live interview platforms and real-time collaboration
Online Coding Platforms	Sharma et al. (2020)	Found that platforms with integrated coding environments improve candidate performance and allow better assessment of problem-solving skills.
Collaborative Systems using WebSockets	Gupta et al. (2021)	Showed that Socket.IO enables efficient real-time data collaborate simultaneously in applications like live coding editors.
AI-Assisted Evaluation Systems	Kumar et al. (2023)	Explored the use of AI in interview systems and found that AI-based insights can reduce bias and improve the accuracy of candidate evaluation.
Remote Hiring Platforms	Patel & Mehta (2022)	Studied modern hiring tools and concluded that integrat efficiency and user experience compared to fragmented systems.

Table 2.1. Research and Case Studies

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2.2 Gap Identification and Solution

Despite advancements in remote interview tools, existing platforms remain fragmented, limited in functionality, and not specifically designed for technical hiring. Many tools focus on only one aspect such as video communication or coding assessment, making the overall interview process inefficient and complex. The following analysis highlights the key gaps and how EmberIQ addresses them.

Table 2.2 - Limitations and Solutions

Identified Gap	Challenges in Current Systems	How Securo Addresses Them
No Unified Interview Platform	Interviewers must use multiple tools like Zoom, coding platforms, and separate evaluation systems with no integration.	EmberIQ integrates video communication, real-time coding, and evaluation into a single unified dashboard.
Lack of Real-Time Coding Collaboration	Many platforms do not support live coding with direct interaction between interviewer and candidate	EmberIQ provides a real-time collaborative code editor where interviewers can observe and interact instantly.
Manualand Inefficient Evaluation	Feedback and scoring are often done outside the platform, leading to inconsistency and delays.	EmberIQ offers structured scorecards, real-time notes, and centralized feedback storage for better decisions.
Limited Security and Privacy	Existing tools may not properly secure interview sessions and candidate data.	EmberIQ ensures secure authentication, encrypted sessions, and role-based access control
Poor Candidate Experience	Candidates face unclear workflows, technical issues, and platform switching during interviews.	EmberIQ provides a smooth, user-friendly interface with integrated features for a better interview experience.

CHAPTER 3

PROBLEM STATEMENT

3.1 Problem Description

In an era where remote hiring has become the standard in the technology industry, organizations face significant challenges in conducting efficient and structured technical interviews. Companies rely on multiple disconnected tools such as video conferencing platforms, coding assessment tools, and manual evaluation systems, which creates complexity, inefficiency, and poor coordination during the interview process. At the same time, candidates often experience unclear workflows, technical issues, and inconsistent interview formats, making it difficult to perform at their best.

The core problem EmberIQ addresses can be decomposed into several interrelated issues in current remote interview practices.

Key Issues in Existing Personal Data Security Practice

- **Fragmented and Uncoordinated Interview Process:** Most organizations rely on multiple platforms such as video conferencing tools, coding environments, and separate evaluation systems, which creates confusion and inefficiency. Interviewers have to switch between tools during the interview, disrupting the flow and reducing the overall effectiveness of candidate assessment
- **Lack of Real-Time Coding and Skill Assessment:** Many existing platforms do not provide proper real-time coding environments where candidates can write, test, and explain their code during interviews. This makes it difficult for interviewers to accurately assess problem-solving skills and technical abilities, leading to incomplete or less reliable evaluation of candidates.
- **Communication and Interaction Experience:**Poor In many existing systems, communication between interviewer and candidate is limited to basic video or chat tools without proper integration. This lack of seamless interaction affects the clarity of problem discussion, reduces engagement, and makes it harder to conduct effective technical interviews.

- **Lack of Pre-Interview Code Monitoring and Control:** In many existing systems, there is no mechanism to monitor or guide candidates before and during coding tasks effectively. Interviewers cannot track candidate actions in real time or ensure proper coding practices, which may lead to unfair assessments or missed evaluation of actual skills.

3.2 Objectives

3.2.1 Specific Goals

The primary objective of this project is to develop a full-stack web application that provides a unified, efficient, and user-friendly platform for conducting remote technical interviews. The system aims to enable interviewers and candidates to communicate, collaborate on coding tasks, and perform structured evaluations within a single integrated environment.

- **Build a Real-Time Interview System:** Implement video and audio communication using WebRTC to enable smooth and secure live interviews between interviewer and candidate.
- **Develop a Real-Time Collaborative Code Editor:** Create a coding environment that supports multiple programming languages, syntax highlighting, and live interaction where interviewers can observe and guide candidates.
- **Implement an Interview Management System:** Design a dashboard to schedule interviews, assign interviewers, and manage interview workflows in a structured manner.
- **Develop a Candidate Evaluation System:** Provide scorecards, real-time notes, and centralized feedback storage to help interviewers assess candidates effectively.

3.2.2 Measurable and Achievable Targets

It outlines measurable and achievable targets for the system, including implementing real-time video interviews using WebRTC, enabling live coding with instant updates, and providing a centralized dashboard for interview management. It also focuses on structured candidate evaluation, basic AI-assisted insights, and secure authentication using JWT to ensure efficient, accurate, and secure interview processes.

Table 3.1 Target Scope

Objective	Measurable Target	Expected Outcome
Live Interview System	Establish real-time video and audio communication with minimal latency using WebRTC.	Interviewers and candidates can conduct smooth and uninterrupted interviews.
Real-Time Code Editor	Enable live coding with instant updates and multi-language support during in terviews.	Interviewers can observe and interact with candidate coding in real time.
Interview Management	Allow scheduling and management of interviews through a centralized dashboard.	Organized interview process with better tracking and coordination.
Candidate Evaluation	Provide scorecards and real-time feedback recording during interviews.	Accurate and structured assessment of candidate performance.
AI Assistance	Generate basic coding suggestions and insights during interviews.	Improved decision-making and reduced evaluation bias.
Security System	Implement secure authentication and role-based access using JWT.	Protection of user data and secure access to the platform.

CHAPTER 4

METHODOLOGY

4.1 Methods and Techniques

4.1.1 Architectural Approach

EmberIQ follows a three-tier architecture with a clear separation between the presentation layer, application layer, and data layer. This design ensures better modularity, scalability, and easy maintenance of the system.

- **Frontend Layer (Presentation):** Built using React.js and Tailwind CSS to create a responsive and user-friendly interface. WebRTC is used for video and audio communication, while Socket.IO enables real-time updates for coding collaboration and chat. The frontend handles all user interactions and displays real-time data.
- **Backend Layer (Application Logic):** Developed using Node.js and Express.js, the backend manages API requests, authentication, interview sessions, and real-time communication. Socket.IO is used for handling live coding and messaging, while JWT is used for secure authentication and session management.
- **Data Layer (Storage):** MongoDB is used as the primary database to store user data, interview records, coding responses, and feedback. It ensures efficient data handling and scalability for large numbers of users.

4.1.2 Security Architecture

Security in EmberIQ is implemented as a layered approach to ensure protection of user data and interview sessions.

- **Transport Security:** All communication between client and server is secured using HTTPS to protect data during transmission.
- **Authentication Layer:** JWT-based authentication is used to verify users, along with role-based access control for interviewers, candidates, and administrators.
- **Data Protection:** Sensitive data such as user credentials are encrypted using hashing techniques like bcrypt, ensuring secure storage.
- **Application Security Layer:** Input validation, secure APIs, and controlled access prevent unauthorized usage, data breaches, and system misuse.

4.1.3 Real-Time Communication and API Integration

EmberIQ implements real-time communication and API integration to ensure smooth interaction between interviewers and candidates during live sessions. The system uses WebRTC for video and audio communication, enabling low-latency and secure peer-to-peer connections without relying on external platforms. This ensures a seamless interview experience.

For real-time coding collaboration and messaging, Socket.IO is used to establish a continuous connection between client and server. It allows instant data synchronization so that any code changes made by the candidate are immediately visible to the interviewer. This ensures effective monitoring and interaction during coding tasks.

4.2 Tools and Software

This section outlines the key technologies used in the system, where React.js and Tailwind CSS are utilized to build a responsive and interactive frontend. WebRTC and Socket.IO enable real-time communication and collaboration, while Node.js with Express.js handles backend operations. Security is ensured through JWT authentication and bcryptjs for password encryption, with MongoDB used for data storage and dotenv for managing environment variables securely.

Table 4.1 Technology Stack

Component	Technology Used	Purpose
Frontend Framework	React.js	Builds interactive and component-based user interface for the platform.
Styling	Tailwind CSS	Provides responsive and modern UI design with faster styling.
Real-Time Communication	WebRTC	Enables live video and audio communication during interviews.
Real-Time Collaboration	Socket.IO	Handles real-time coding updates and chat between users.
Backend Framework	Node.js + Express.js	Manages server-side logic, APIs, and application flow.
Authentication	JWT	Ensures secure user login and session management.
Password Security	bcryptjs	Encrypts user passwords for secure storage.

Database	MongoDB	Stores user data, interview records, and feedback efficiently.
Environment Management	dotenv	Manages environment variables securely.
API Handling	CORS, cookies	Handles cross-origin requests and session data.
File Storage	Cloudinary / AWS S3	Stores media files and documents if required.
Development Tool	Nodemon	Automatically restarts server during development.
Containerization (Optional)	Docker	Provides scalable and consistent deployment environment.
Version Control	Git + GitHub	Manages code versions and supports collaboration.
Deployment	AWS / Vercel / GCP	Hosts the application with scalability and reliability.

4.3 Data Collection and Analysis Procedures

4.3.1 Interview Data Collection

When a user schedules or participates in an interview, the system collects relevant data such as user details, interview session information, coding activity, chat messages, and evaluation feedback through backend APIs. This data is securely transmitted to the server using Node.js and Express.js and stored in MongoDB for further processing and analysis.

During the interview, real-time data such as code changes, typing activity, and communication between interviewer and candidate is captured using Socket.IO. This ensures that all interactions are recorded and can be used for monitoring and reviewing the candidate's performance. Interviewers can also add real-time notes, assign scores, and evaluate different aspects such as problem-solving ability, coding skills, and communication.

The system organizes this collected data under each candidate's profile, allowing easy access and structured storage of interview history. This helps in maintaining consistency in evaluation and enables comparison between different candidates.

The collected data is further analyzed to generate meaningful insights, such as coding efficiency,

accuracy, time taken to solve problems, and overall interview performance. These insights assist recruiters in making informed and data-driven hiring decisions.

Finally, the frontend presents this analyzed data through dashboards and structured views, making it easy for interviewers and administrators to review candidate performance and track interview outcomes effectively.

4.3.2 Interview Data Handling

All interview-related data in EmberIQ is handled in a structured and secure manner to ensure consistency and reliability. When an interview session is conducted, data such as video session details, coding activity, chat messages, and evaluation inputs are collected and processed in real time. The system captures coding data through the real-time editor and stores it along with timestamps, allowing interviewers to review the candidate's approach and problem-solving steps. Chat messages and communication logs are also recorded to maintain a complete interaction history.

All collected data is securely transmitted to the backend and stored in MongoDB with proper organization under each user and interview session. Sensitive information such as user credentials is protected using encryption techniques, and access to data is controlled through role-based authentication.

The server ensures that only authorized users can access interview data, and no unnecessary or sensitive information is exposed. This structured data handling helps maintain transparency, security, and efficient data management throughout the interview process.

4.3.3 AI Context Analysis

EmberIQ incorporates AI-based analysis to assist in evaluating candidate performance and improving decision-making. When an interviewer interacts with the AI system or requests insights, the platform retrieves relevant data such as candidate performance, coding activity, and evaluation scores from the database.

This information is processed and provided as context to the AI system, which then generates meaningful suggestions, feedback, or analysis based on the candidate's performance. The AI can help identify strengths, weaknesses, and patterns in coding behavior.

The system ensures that AI responses are based on actual interview data rather than generic outputs, making the analysis more accurate and useful. The results are presented in a clear and structured format on the frontend, helping interviewers understand insights easily.

This approach enhances the evaluation process by combining human judgment with intelligent data-driven analysis, leading to better and more informed hiring decisions.

CHAPTER 5

SYSTEM DESIGN

5.1 Architectural Design

EmberIQ is built on a modular, component-based full-stack architecture using modern web technologies. The system separates the frontend, backend, and database layers to ensure scalability, maintainability, and efficient performance. This architecture allows smooth communication between components while keeping the system organized and easy to manage.

5.1.1 Three-Tier Architecture Overview

- **Presentation Layer:** Built using React.js and styled with Tailwind CSS, this layer handles all user interactions and interface rendering. It includes features like video interface, coding editor, dashboards, and real-time updates for a smooth user experience.
- **Application Layer:** The backend is developed using Node.js and Express.js, which manages API requests, authentication, interview sessions, and real-time communication. Socket.IO is used to handle live coding and messaging, while JWT ensures secure user authentication.
- **Data Layer:** MongoDB Atlas stores user profiles, interview data calling response and feedback . It povide efficient data management and supports for handling multiple user sessions

5.1.2 Security Architecture Layers

Security in EmberIQ is implemented using a layered approach to ensure that failure of one layer does not compromise the entire system. Each layer provides a specific level of protection for user data and system functionality.

Layer	Components	Protection Provided
Layer 1: Transport & Storage	HTTPS protocol, secure database (MongoDB)	Ensures that all data transmitted between client and server is encrypted and stored securely.
Layer 2: Data Protection	bcrypt password hashing, secure data handling	Protects sensitive user data such as passwords and personal information from unauthorized access.

Layer 3: Authentication	JWT-based authentication, role-based access control	Ensures that only authorized users (admin, interviewer, candidate) can access the system.

5.2 Detailed Design of Modules and Components

5.2.1 Interview Module

The Interview Module is the core component of EmberIQ and acts as the primary interaction layer between interviewer and candidate. It enables real-time communication, coding collaboration, and smooth interview flow within a single interface. The module supports live sessions and ensures all activities are synchronized and recorded for evaluation purposes.

- **Video and Audio Communication:** The system uses WebRTC to enable real-time video and audio interaction between interviewer and candidate. It provides a stable and low-latency communication channel for conducting interviews.
- **Real-Time Coding Environment:** A collaborative code editor allows candidates to write and execute code while interviewers can observe changes instantly. It supports multiple programming languages and includes syntax highlighting.
- **Live Interaction and Monitoring:** Interviewers can track candidate activity in real time, provide guidance, and communicate through built-in chat. This ensures better understanding of the candidate's approach and problem-solving skills.

5.2.2 Real-Time Code Editor Module

The Real-Time Code Editor Module is a key component of EmberIQ that enables collaborative coding between interviewer and candidate during live interviews. It provides an interactive environment where coding can be performed, observed, and evaluated in real time.

- **Code Editing and Execution:** The system allows candidates to write code in a structured editor with support for multiple programming languages. It includes features like syntax highlighting and optional code execution for better understanding of logic.
- **Real-Time Synchronization:** Using Socket.IO, all code changes are instantly synchronized between interviewer and candidate. This ensures that both users see the same code simultaneously without delay.
- **Monitoring and Interaction:** Interviewers can closely observe the candidate's coding

process in real time, gaining insight into their problem-solving approach, logic, and coding style. They can actively guide candidates during the problem-solving phase by offering hints, clarifying requirements, or suggesting alternative approaches when needed. Additionally, interviewers can provide instant feedback through chat or live discussion, enabling a more interactive and collaborative evaluation experience while helping candidates understand their strengths and areas for improvement.

5.2.3 Candidate Evaluation Module

The Candidate Evaluation Module is designed to assess the performance of candidates during and after the interview process. It provides a structured and efficient way for interviewers to record observations, assign scores, and make informed hiring decisions.

- **Scorecard System:** The platform provides predefined evaluation criteria where interviewers can rate candidates on technical skills, problem-solving ability, and communication. This ensures a standardized assessment process.
- **Real-Time Notes:** Interviewers can add notes during the interview session, capturing important observations about the candidate's approach, behavior, and coding style.
- **Performance Analysis:** The system organizes evaluation data and presents it in a structured format, helping interviewers compare candidates and analyze their strengths and weaknesses.
- **Centralized Feedback Storage: All** feedback and evolution records stored in database .

5.2.4 Communication and Chat Module

The Communication and Chat Module enable smooth interaction between interviewer and candidate during the interview session. It ensures clear communication and supports real-time exchange of messages alongside video and coding features.

- **Real-Time Messaging:** The system provides an integrated chat feature where interviewers and candidates can send and receive messages instantly during the interview
- **Seamless Integration:** Chat is integrated with the video and coding interface, allowing users to communicate without switching between different platforms.
- **Live Updates:** Using Socket.IO, messages are delivered in real time without delay,

ensuring continuous and effective communication throughout the session.

- **Session Management:** Chat history is maintained for each interview session, allowing interviewers to review conversations and maintain proper records if needed.

5.2.5 AI-Assisted Interview Module

The AI-Assisted Interview Module provides intelligent support during the interview process by offering insights, suggestions, and analysis based on candidate performance.

- **Context-Based Analysis:** The system collects relevant data such as coding activity, performance scores, and interviewer feedback from the database and uses it as input for AI processing to generate meaningful insights.
- **Smart Suggestions:** The AI provides coding hints, error detection, and improvement suggestions to assist interviewers in understanding the candidate's approach and problem-solving ability.
- **Performance Insights:** AI analyzes candidate behavior, coding efficiency, and accuracy to highlight strengths and weaknesses, helping interviewers make better and unbiased decisions

5.2.6 Additional Feature Modules

- **Interview Scheduling Module:** Allows users to schedule interviews, assign interviewers, and manage interview timings through a centralized dashboard. It ensures a structured and organized interview process.
- **Screen Sharing Feature:** Enables candidates to share their screen during the interview for better explanation of solutions, walkthroughs, or demonstrations.
- **Role-Based Access Control:** Provides different access levels for admin, interviewer, and candidate to ensure secure and controlled usage of the platform.
- **Session Recording and History:** Stores interview session details including coding activity, chat, and feedback for future reference and analysis.
- **Notification System:** Sends alerts and updates related to interview schedules, session status, and important activities to keep users informed

5.2 Diagram

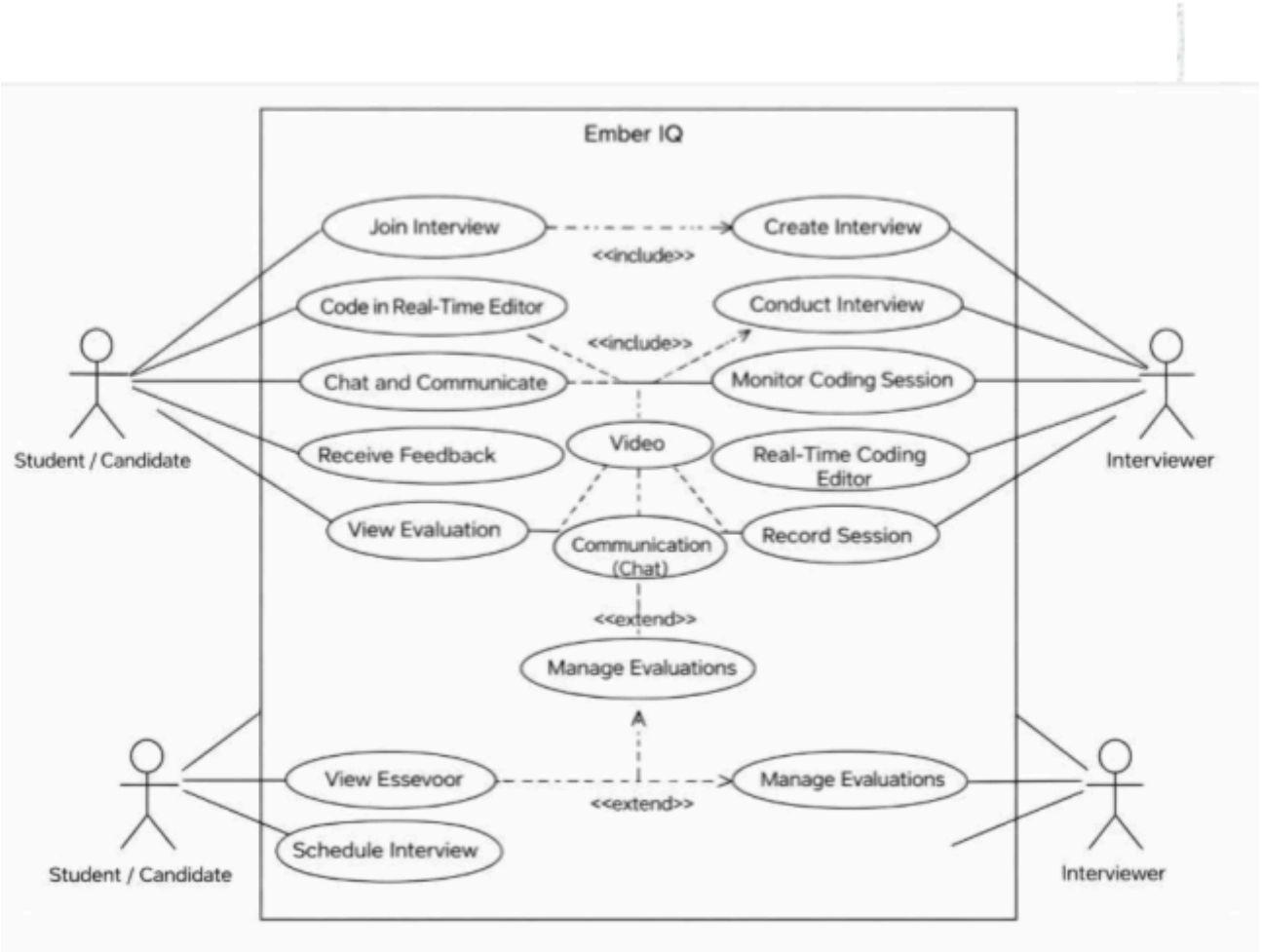

Figure 5.3.1: Use Case Diagram

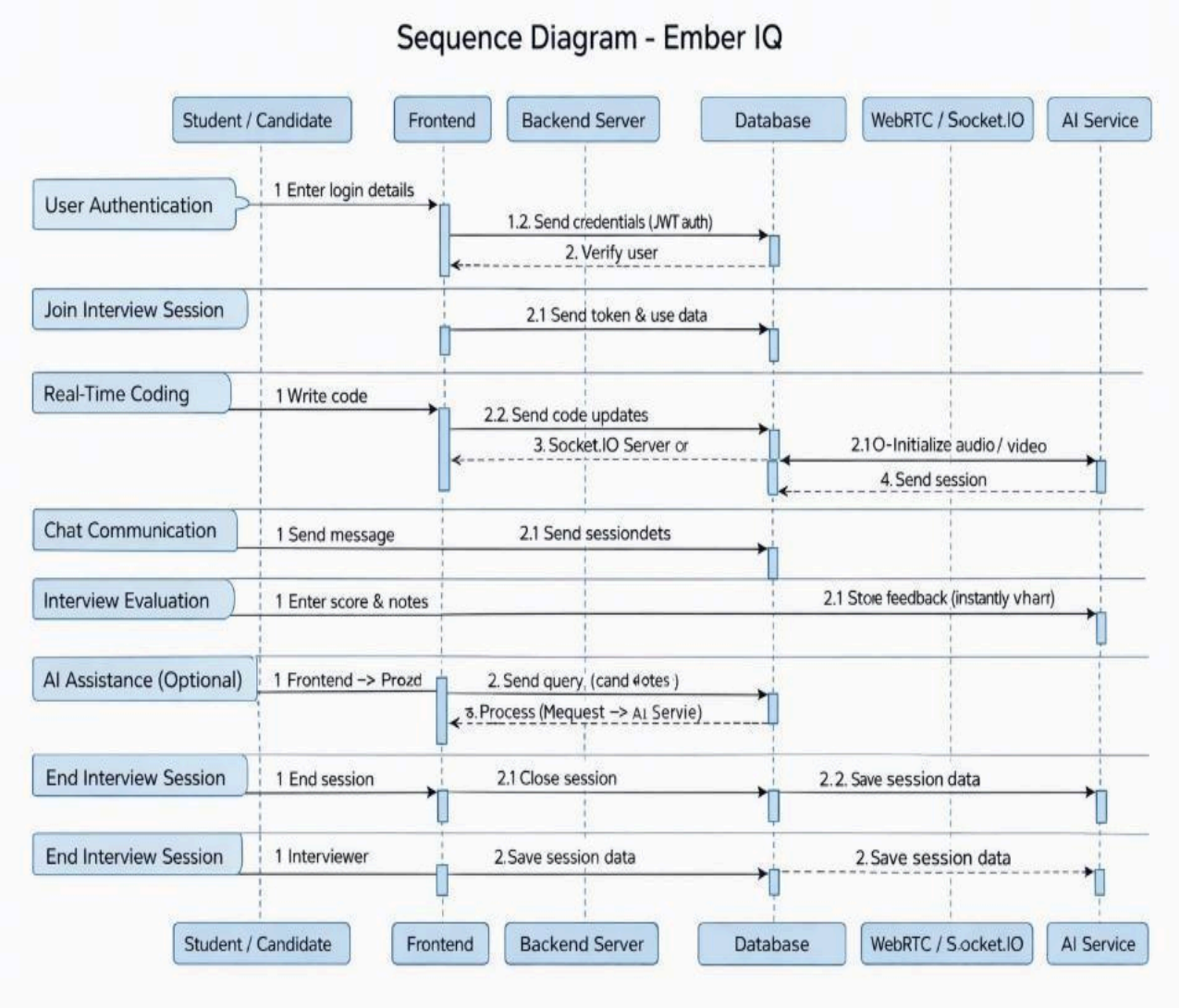

Figure 5.3.2: Sequence Diagram

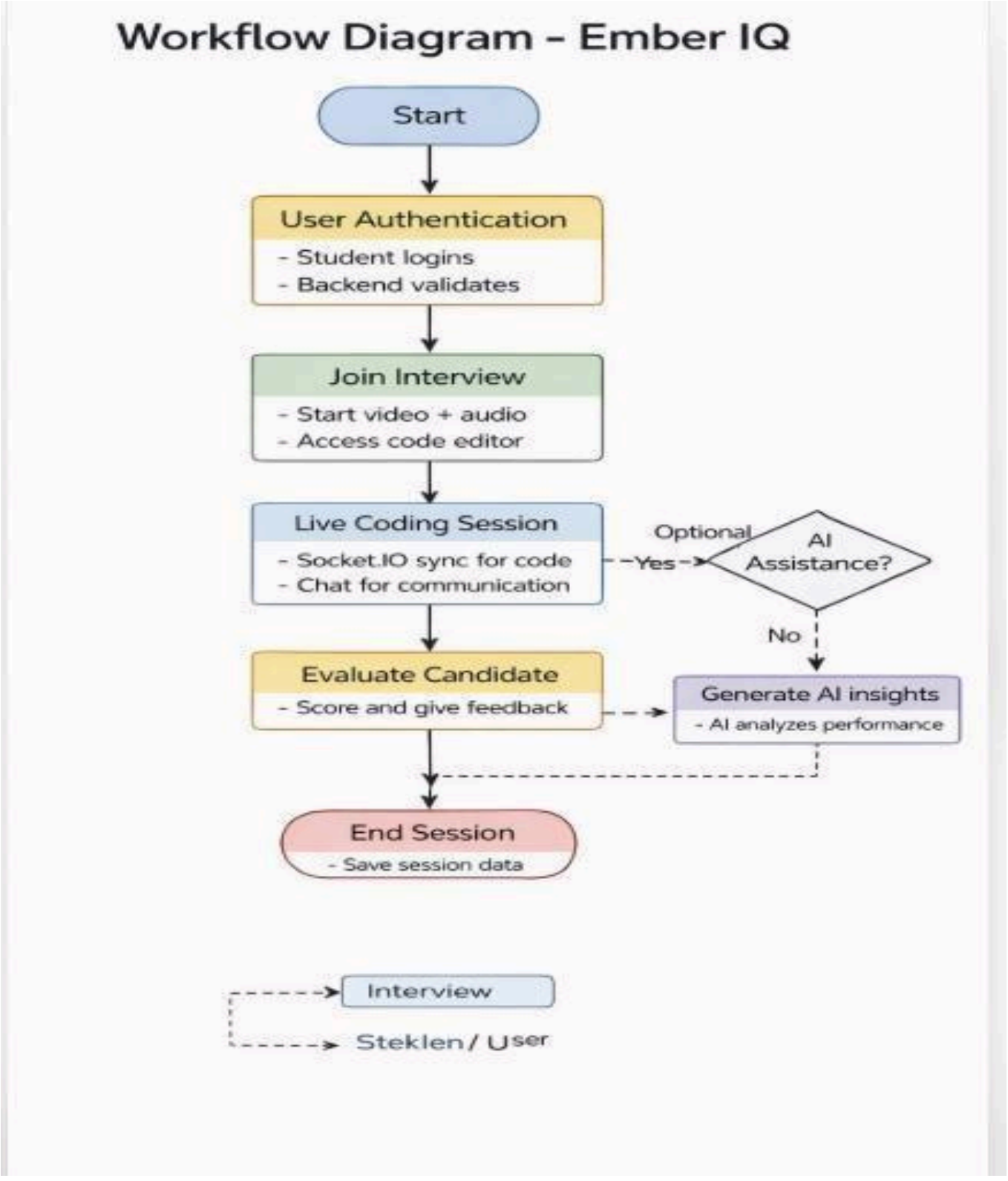

Figure 5.3.3: Workflow Diagram

CHAPTER 6

IMPLEMENTATION

6.1 Implementation Process

The implementation of EmberIQ was carried out in a phased approach, focusing first on core interview functionalities and then integrating additional features to enhance usability and performance. The development followed a modular approach, where each component was built and tested separately before integrating into the complete system.

Implementation Phases

- **Foundation and Authentication:** Initial setup of the project using React.js for frontend and Node.js with Express.js for backend. Tailwind CSS was integrated for UI design, and JWT-based authentication was implemented for secure login and session management. MongoDB was connected using Mongoose for database operations
- **Core Interview Features:** Development of the main interview system including WebRTC-based video and audio communication, real-time code editor, and Socket.IO integration for live collaboration and chat functionality.
- **Interview Management:** Implementation of features like interview scheduling, session handling, and dashboard for managing interview workflows and participants.
- **Evaluation System:** Development of scorecards, real-time note-taking, and centralized feedback storage to support structured candidate evaluation.
- **AI Integration:** Integration of AI features to provide coding suggestions, performance insights, and intelligent assistance during interviews.
- **Testing and Optimization:** Final phase included debugging, performance optimization, UI improvements, responsiveness, and thorough testing to ensure system stability and smooth user experience.

6.1.1 Frontend Architecture

The frontend of EmberIQ is organized using a modular and feature-based structure to ensure scalability and maintainability. It is developed using React.js, where each major feature such as interview, coding editor, and dashboard is divided into separate components and folders.

Each module contains its own components, hooks, and utility functions, making the code easy to manage and update.

6.1.2 Backend Architecture

The backend of EmberIQ is built using Node.js and Express.js to handle all server-side logic and application functionalities. It manages user authentication, interview sessions, real-time communication, and data processing efficiently.

JWT-based authentication is implemented to secure all protected routes, ensuring that only authorized users can access the system. Role-based access control is applied to differentiate permissions for candidates, interviewers, and administrators.

All API requests are handled through structured endpoints, which process data between the frontend and the database. Sensitive operations and data handling are performed on the server side to maintain security and prevent exposure of critical information.

6.2 Integration of Different Modules and Components

6.2.1 Frontend and Backend Integration

The frontend of EmberIQ communicates with the backend using REST APIs through the Fetch API. React components send requests to the Node.js and Express.js server to handle operations like authentication, interview management, and data retrieval. Real-time features such as coding updates and chat are managed using Socket.IO, ensuring instant communication between client and server. This integration allows smooth data flow and a responsive user experience.

6.2.2 Authentication Integration

Authentication in EmberIQ is handled using JWT-based security. Users register and log in through secure endpoints, and a token is generated for session management. This token is used to access protected routes within the application. Role-based access control ensures that different users such as candidates, interviewers, and admins have appropriate permissions. This integration maintains security and prevents unauthorized access.

6.2.3 Real-Time Communication Integration

The system integrates WebRTC and Socket.IO to enable real-time features. WebRTC is used for video and audio communication, allowing live interviews, while Socket.IO handles real-time data transfer such as code synchronization and chat messages. These technologies work together to provide seamless interaction between users during the interview process.

6.2.4 Data Base Integration

MongoDB is integrated with the backend using Mongoose to manage data storage and retrieval. All data related to users, interviews, coding sessions, and evaluations is stored in structured collections. This ensures efficient data handling, consistency, and scalability of the system

6.2.5 Module Integration

All modules such as interview system, code editor, chat, evaluation, and AI assistance are interconnected through the backend. Each module communicates with others through APIs and shared data, creating a unified platform. This integration ensures that all features work together smoothly within a single system.

CHAPTER 7

TESTING

Testing: The testing phase ensured that EmberIQ functions correctly across all modules, with particular emphasis on critical components such as real-time communication, authentication flow, and data handling. A multi-layered testing strategy was employed to validate functionality, performance, and system reliability before deployment.

7.1 Testing Strategies and Methodologies

7.1.1 Unit Testing

Purpose: Validates that individual components and functions of EmberIQ produce correct outputs for given inputs.

- **Authentication Module**: Verified that login and registration functions correctly validate user credentials and generate valid JWT tokens while handling invalid inputs properly.
- **Real -Time Communication**: Tested WebRTC and Socket.IO connections to ensure stable video/audio communication and accurate real-time data transfer.
- **Code Editor Functionality:** Confirmed that the code editor correctly handles input, syntax highlighting, and real-time synchronization between users.
- **Evaluation Module**: Verified that scorecards, notes, and feedback are correctly recorded and stored in the database without data loss.

7.1.2 Integration Testing

Purpose: Verifies that system components interact correctly across API boundaries, database operations, and external service integrations.

- **Authentication flow:** End-to-end testing of Clerk sign-up, sign-in, session persistence, and protected route enforcement.
- **Vault encryption roundtrip:** Full pipeline testing from client-side encryption through API upload to MongoDB storage and back through decryption to confirm zero data loss or corruption.
- **Breach monitoring pipeline:** Verified that XposedOrNot API responses are correctly parsed, persisted to MongoDB, and rendered through Recharts components.

7.1.3 Security Testing

Purpose: Verifies that different components of EmberIQ work together correctly across APIs, database operations, and real-time services.

- **Authentication Flow:** End-to-end testing of user registration, login, JWT token generation, session persistence, and access to protected routes.
- **Real-Time Communication Integration**: Tested WebRTC and Socket.IO integration to ensure smooth video/audio communication and real-time synchronization of code and chat messages.
- **Interview Data Flow:** Verified that interview data such as coding activity, chat messages, and evaluation feedback is correctly sent from frontend to backend and stored in MongoDB without data loss.

7.1.4 User Acceptance Testing

Testing sessions were conducted with students and professionals from both technical and non-technical backgrounds. Users were given access to the EmberIQ platform and asked to perform tasks such as joining interviews, coding, and interacting with features without guidance. Feedback was collected on usability, clarity, performance, and overall user experience to identify areas of improvement and ensure the system meets real-world requirements.

7.2 Test Cases and Results

This section presents the test cases and their results, showing that all core functionalities performed as expected. User registration, protected routes, and role-based access worked correctly, ensuring proper authentication and security. Real-time interview sessions, code synchronization, and chat features were also successfully tested, demonstrating stable communication and seamless real-time interaction across the platform.

Table 7.2.1: Authentication and Access Control Test Cases

Test Case	Description	Expected Outcome	Actual Outcome	Status
User Registration	Register new user using email/password	Account created and stored in database	Users registered successfully	Passed

Protected Route	Unauthenticated access to dashboard	Redirect to login page	Redirected correctly	Passed
Role-Based Access	Unauthorized user access	Access denied	Access restricted correctly	Passed
Real-Time Interview	Start interview session	Stable video/audio connection	Connection successful	Passed
Code Sync	Write code during interview	Real-time updates	Sync working correctly	Passed
Chat System	Send message	Instant delivery	Messages received instantly	Passed

Table 7.2.2: Feature Functionality and API Integration Test Case

Test Case	Description	Expected Outcome	Actual Outcome	Status
Interview Start	Join interview session	Video/audio starts successfully	Session started smoothly	Passed
Code Execution	Write and run code	Code executes correctly	Output generated correctly	Passed
Evaluation Save	Submit feedback and score	Data stored in database	Data saved successfully	Passed
Chat System	Send message during interview	Instant message delivery	Message delivered instantly	Passed
AI Assistance	Request AI help	Relevant suggestions generated	AI responded correctly	Passed

7.3 Bug Fixes and Improvements

7.3.1 Bug: Video Call Connection Failure

Issue: In some cases, WebRTC failed to establish a stable video/audio connection due to network inconsistencies.

Fix: Improved connection handling by adding fallback ICE servers and retry mechanisms.

Result: Video and audio communication now works smoothly across different networks and devices.

7.3.2 Bug: Real-Time Code Sync Delay

Issue: Code updates were sometimes delayed due to improper Socket.IO event handling, causing mismatch between interviewer and candidate screens. **Fix:** Optimized event emission and implemented efficient state synchronization.

Result: Code updates are now reflected instantly with no noticeable delay.

7.3.3 Bug: AI Suggestion Inaccuracy

Issue: AI module sometimes provided irrelevant or generic suggestions due to lack of proper context from interview data.

Fix: Improved input data handling by passing structured candidate performance and coding data to the AI module.

Result: AI now generates more relevant and context-aware suggestions.

7.3.4 Bug: Interview Session Lag

Issue: Users with multiple monitored emails triggered rate limiting on simultaneous breach checks.

Fix: Implemented a sequential queuing system with 500ms delays between API calls and a server-side cache layer that serves stored breach data for recently checked emails.

Result: Multi-email monitoring operates reliably without rate limit errors.

7.3.5 Bug: Theme Preferences Not Persisted Across Sessions

Issue: Custom theme settings were stored only in React state and lost on page reload.

Fix: Migrated theme storage to local Storage with a custom useTheme hook that reads from local Storage on mount and persists changes on update.

Result: Theme preferences are correctly restored on all subsequent sessions.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Results Obtained

The implementation of EmberIQ successfully delivered all planned core features required for a remote technical interview platform. The system operates as a complete and functional web application with a user-friendly interface, real-time communication, and integrated interview tools.

User Interface and System Functionality

The platform provides a clean landing page that introduces the purpose of EmberIQ with a clear call-to-action for users to start or join interviews. The main dashboard includes a structured layout with access to key modules such as interview sessions, code editor, chat system, and evaluation tools.

The interview module provides a smooth experience with real-time video and audio communication using WebRTC. Users can join sessions easily, and the system maintains stable connectivity throughout the interview.

The real-time code editor is one of the most important features, allowing candidates to write code while interviewers can observe changes instantly. It supports multiple languages and ensures proper synchronization using Socket.IO.

The evaluation module allows interviewers to give scores, add notes, and store feedback in a centralized system. This helps in maintaining structured and consistent candidate assessment.

The AI-assisted feature enhances the platform by providing coding suggestions and performance insights based on candidate activity. It helps interviewers understand the candidate's strengths and weaknesses more effectively.

Overall, EmberIQ provides a seamless and efficient interview experience by integrating communication, coding, and evaluation into a single platform.

UI Snapshots

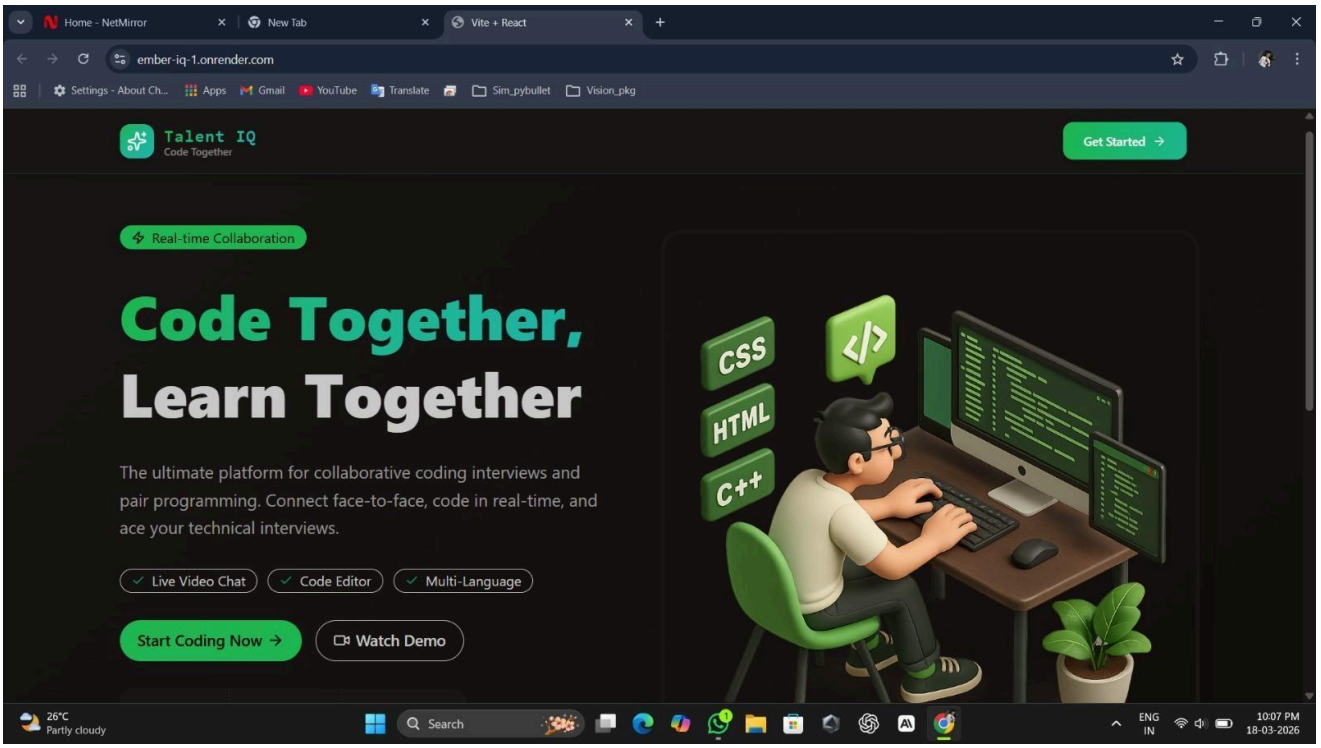

Figure 8.1.1: Landing / Home Screen

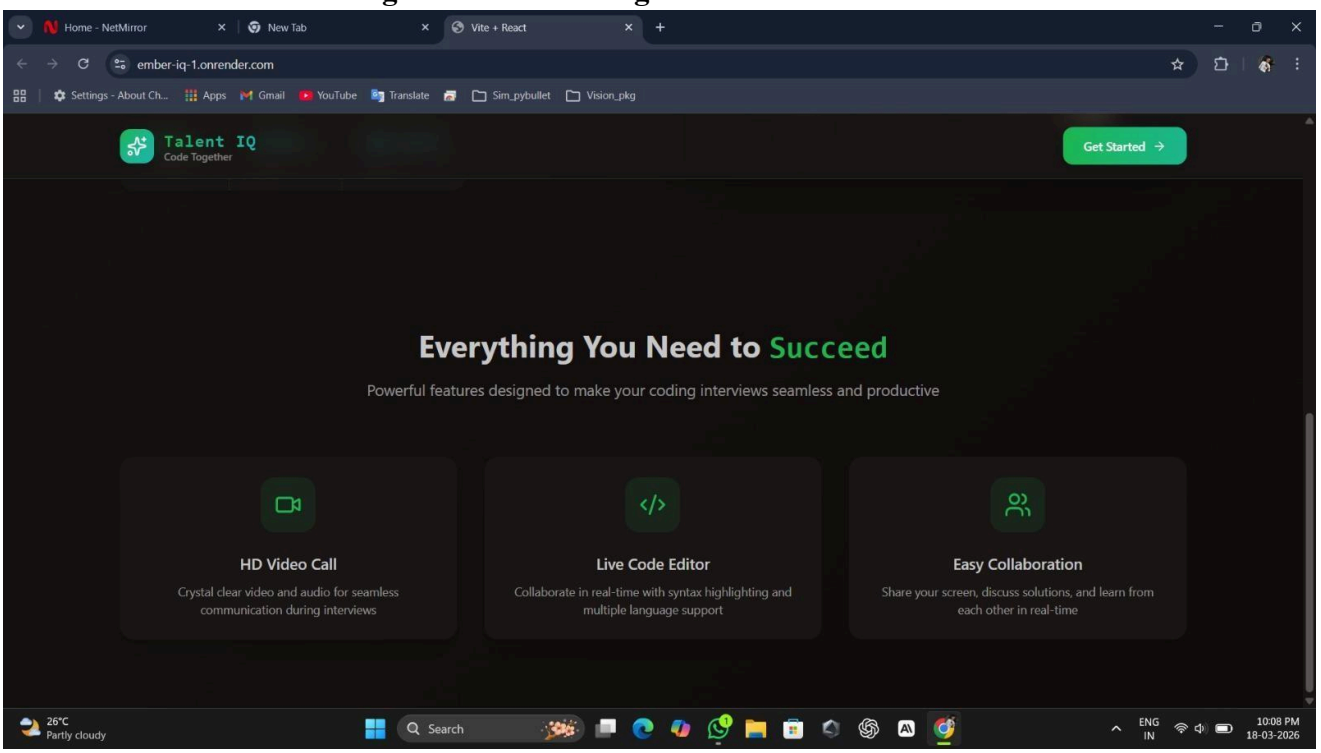

Figure 8.1.2:

8.2 Analysis and Interpretation of the Results

8.2.1 System Performance Verification

The system performance of EmberIQ was verified through multiple test scenarios. Real-time features such as video communication and code synchronization were tested under different network conditions, and the platform maintained stable performance with minimal latency. Data storage and retrieval from MongoDB were also validated to ensure accuracy and consistency.

8.2.2 Integration Accuracy

All integrated technologies such as WebRTC, Socket.IO, and backend APIs were tested to ensure smooth interaction between modules. Video/audio communication worked reliably, real-time code updates were synchronized correctly, and data exchange between frontend and backend was accurate. The system showed consistent performance across different sessions.

8.2.3 User Acceptance Feedback

User testing indicated a high level of satisfaction with the platform’s usability and functionality. Users found the interface easy to understand and appreciated the integration of video, coding, and evaluation in one place. The real-time coding and communication features were especially well received. Some users suggested improvements such as better onboarding guidance and enhanced UI clarity for first-time users.

8.3 Comparison with Expected Outcomes

The comparison between expected and actual outcomes demonstrates that the system has successfully met its intended objectives across all major features. Each module, including live interviews, real-time code collaboration, interview management, candidate evaluation, AI assistance, and security, performed as anticipated with stable and efficient results. Minor variations were negligible, and overall performance aligned well with expectations, indicating a robust and reliable implementation of the proposed system.

Table 8.3.1: Outcome Comparison

Feature	Expected Outcome	Actual Outcome	Remarks
Live Interview	Smooth video/audio communication	Stable connection with low latency	Meets expectations
Real-Time Code Editor	Instant code collaboration	Code synced in real time using Socket.IO	Meets expectations
Interview Management	Easy scheduling and tracking	Dashboard manages sessions effectively	Meets expectations
Candidate Evaluation	Structured scoring and feedback	Scorecards and notes stored correctly	Meets expectations
AI Assistance	Basic coding suggestions and insights	AI provides useful performance insights	Meets expectations
Security System	Secure authentication and data protection	JWT and role-based access working properly	Meets expectations

CHAPTER 9

CONCLUSION & FUTURE SCOPE

9.1 Findings and Significance

EmberIQ successfully demonstrates that a complete and efficient remote technical interview platform can be developed as a unified web-based solution without compromising performance or user experience. The project proves that integrating real-time communication, collaborative coding, and structured evaluation into a single system is both technically feasible and practically useful.

Key findings from the project include the effectiveness of real-time technologies such as WebRTC and Socket.IO in enabling smooth video communication and live coding collaboration. The system also validates that structured evaluation through scorecards and centralized feedback improves the accuracy and consistency of hiring decisions.

The project highlights the importance of user experience, showing that a simple and well-organized interface can significantly improve both interviewer and candidate performance. The integration of AI assistance further enhances the platform by providing insights and support during interviews.

Overall, EmberIQ demonstrates that the challenges of remote technical hiring can be addressed through a well-designed and integrated system. It contributes to modern recruitment by making the interview process more efficient, structured, and accessible for organizations and candidates.

9.2 Achievements and Contributions

- **Unified Interview Platform:** Successfully integrated video communication, real-time coding, chat, and evaluation into a single platform, eliminating the need for multiple tools during interviews.
- **Real-Time Collaboration System:** Implemented WebRTC and Socket.IO to enable live video interaction and instant code synchronization between interviewer and candidate.
- **Structured Evaluation System:** Developed scorecards, real-time notes, and centralized feedback storage to improve accuracy and consistency in candidate assessment.
- **AI-Assisted Interview Support:** Integrated AI features to provide coding suggestions, performance insights, and assist interviewers in making better decisions.
- **Secure and Scalable Architecture:** Implemented JWT-based authentication, role-based access control, and MongoDB database to ensure secure and scalable system performance.
- **Production-Ready Application:** Built a complete full-stack application using modern technologies like React, Node.js, and MongoDB, suitable for real-world usage and deployment.

9.3 Future Scope and Potential for Further Research

Phase 1: Advanced Features

- AI-based candidate performance prediction to analyze coding behavior and estimate overall interview success.
- Personalized interview analytics dashboard showing performance trends and improvement suggestions.
- Automated feedback generation to assist interviewers with quick and structured evaluation reports.
- Integration of advanced proctoring features to monitor candidate activity and ensure fair assessment.
- Implementation of password less authentication methods like OTP or biometric-based login for enhanced security

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Phase 2: Collaboration and Enterprise

- Team-based interview management system with role-based access control for admins, interviewers, and HR teams.
- Detailed activity logs and audit tracking for all interview sessions and user actions for better monitoring and compliance.
- Compliance and Reporting: Ability to generate detailed reports for interview data, candidate performance, and system usage to support organizational standards and documentation.
- Developer API Integration: Provide APIs that allow third-party platforms to integrate EmberIQ features such as interview scheduling, coding sessions, and evaluation systems programmatically.

Phase 3: Platform Expansion

- **Browser Extension:** Development of a browser extension to support quick interview access, notifications, and real-time assistance during coding tasks.
- **Mobile Application:** Creation of a React Native-based mobile app to allow users to join interviews, receive updates, and manage schedules with features like biometric login.
- **Desktop Application**: Development of a desktop version using Electron or similar frameworks to provide a stable and dedicated environment for interviews and coding.
- **CLI tool for developers:** A command line tool help developers manage interview session.

Research Opportunities

- **AI-Based Interview Analysis**: Research on improving AI models to analyze candidate behavior, coding patterns, and decision-making for more accurate evaluation.
- **Distributed Learning Systems**: Exploration of techniques to train AI models using decentralized data while maintaining user privacy and system efficiency.
- **Usability of Secure Systems**: Study of user interaction and experience with secure interview platforms to improve accessibility and ease of use for non-technical users.

CHAPTER 10

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